

AVICHAL BAWEJA

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SUMMARY

Backend Engineer with strong expertise in building scalable APIs, microservices, and data processing systems. Proficient in Python, FastAPI, and database design with hands-on experience in PostgreSQL, MongoDB, and Redis. Experienced in developing backend services for AI-powered applications and enterprise systems in fast-paced startup environments.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, Node.js, Express.js, REST APIs, GraphQL, Microservices, WebSockets
Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases, Database Optimization, Indexing
Cloud & DevOps: GCP, AWS, Docker, Kubernetes, CI/CD, GitHub Actions, Nginx, Load Balancing, Linux
Programming: Python, Java, JavaScript, TypeScript, C, SQL, Bash

EXPERIENCE

Founding Engineer | *Datavedam*

Oct 2025 – Present

- Designed and developed scalable backend services using Python (FastAPI) for automation, analytics, and communication workflows serving multiple stakeholders.
- Built AI-powered solutions including chatbot systems, automation tools, and data-driven platforms for educational institutions and government bodies.
- Architected multi-tenant systems with modular configurations, jurisdiction-based data handling, and role-based access control.
- Implemented DevOps workflows using Docker ensuring smooth CI/CD pipelines, production reliability, and zero-downtime deployments.
- Optimized system performance through caching strategies, database indexing, and query optimization for concurrent users and real-time requests.

Business Analyst | *Purshotam Profiles Pvt. Ltd.*

Jan 2025 – July 2025

- Identified new business opportunities in the solar projects sector; prepared proposals and performance reports for client engagement and decision-making.
- Collaborated with cross-functional teams to optimize sales processes and strengthen B2B client relationships across verticals.

Computer Vision Intern | *TechEagle Innovations Pvt. Ltd.*

Apr 2024 – May 2024

- Developed a CV-based drone system integrating YOLOv5 with real-time feedback control for stability, autonomous navigation, and obstacle avoidance.
- Designed an autonomous landing system in ROS/Gazebo using Python, OpenCV, and ArUco markers for precise coordinate-based landings.

IT Intern | *Sankalptaru Foundation*

Jan 2023 – Mar 2023

- Built predictive models using Python, Pandas, and scikit-learn for forecasting farm conditions; performed demand analysis with Matplotlib for agroforestry planning.

PROJECTS

Datavedam – AI Powered School ERP Platform | *Python, FastAPI, React, Docker*

- Built RAG pipeline for CBSE circulars with vector embeddings for intelligent querying; developed AI Copilot and OCR/OMR processing pipelines.
- Implemented QR-code based admission workflow handling 1000+ enrollments; integrated multiple APIs for end-to-end school automation.

Manufacturing Dispatch Management Portal | *React, FastAPI, PostgreSQL*

- Designed centralized web portal used by 15+ manufacturing plants for real-time dispatch management, logistics tracking, and GRN reconciliation.

Event Planner | *React, FastAPI, REST API, LLM*

- Built full-stack platform with integrated LLM-powered AI assistant for automated event descriptions, RSVP handling, and content generation.

Handwritten Text to LaTeX Converter | *Python, OCR, Image Processing, ML*

- Built OCR + LLM pipeline to extract handwritten notes and generate accurate LaTeX documents with optimized prompts for math expressions.

EDUCATION

Christ University

B.Tech. Computer Science & Engineering, Honors in AI & ML

Bengaluru, Karnataka

2021 – 2025

Delhi Public School

Senior Secondary School (CBSE)

Agra, Uttar Pradesh

2019 – 2021